FR3 Task Design Document

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1.0.3	Youssef Ezzyat	Updated to reflect decoding/stim on all words	1/8/2016
1.0.2	Michael Kahana	Edited and reorganized document	1/3/2016
1.0.1	Youssef Ezzyat	Revised classifier training process	12/23/2015
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Overview

The FR3 task implements closed loop brain stimulation during a memory task to test the hypothesis that direct electrical stimulation (DES) of the brain during poor memory states results in better memory than a control condition with no brain stimulation. This work builds on two prior experiments: FR1 and FR2.

In FR1 we recorded brain activity from multiple implanted electrodes during the encoding and retrieval of word lists and identified neural correlates of good and poor memory encoding states (henceforth biomarkers). We sought to predict whether a given item would be recalled based on the spectral decomposition of the time-series of bipolar neural recordings from all N channels recorded. We thus fit a L2-regularized logistic regression model to the item level data in which the LHS variable was the recalled/non-recalled status of an item and the RHS variables were the wavelet power in each of 8 frequency bands and across all N channels (henceforth classifier). The model was fit to data from sessions 1 through M-1 and validated on session M, with the penalty parameter determined based on the distribution of best fitting parameters across all patients who performed the experiment excluding the patient whose data were being fit. The model was able to reliably predict recall based on the spectral decomposition of the encoding period data, as described by Ezzyat et al (submitted).

In FR2 we used open loop brain stimulation to assess whether DES would improve or impair memory for the items immediately following the cessation of stimulation. Although we found that DES frequently resulted in impaired memory there were some cases of memory improvement. Using the logistic regression model described above we found that patients for whom DES resulted in improved classifier output (i.e., predicting better memory) exhibited better memory for the post-stimulated items. In contrast, those patients for whom DES resulted in reduced classifier output exhibited worse memory for the post-stimulated item. We further hypothesized that DES is more likely to improve memory when it arrives during poor memory states than when it arrives during good memory states. We thus partitioned items into terciles based on the state of the classifier and found that DES resulted in improved memory when it arrived during the lowest tercile; no such beneficial effects were observed when DES arrived during higher terciles of classifier output.

The present study (FR3) aims to directly test the hypothesis that DES delivered during poor memory states can produce reliably better memory than observed on control lists without any DES. On half of the experiment lists, DES will be under control of an algorithm that evaluates the state of the brain for each presented word and decides whether or not to stimulate vs. control lists in which no stimulation is provided.

Methods

Except as noted, the design of the behavioral task and the procedure for it are identical to FR1. The reader is referred to that design document for further details. Subjects will perform delayed free recall on 26 study-test lists, the first of these being practice (all data from all lists, including the practice list, will be recorded). Stimulation is not delivered to the brain during lists 1-4 as these lists are used to evaluate the ability of the logistic-regression classifier, trained on the

same patient's FR1 data, to predict the recalled status of items in FR3. Following the first 4 lists, the remaining 22 lists will be equally divided into two conditions: Stim and NoStim lists. These 22 lists will be pseudorandomly interleaved to balance the number of Stim and NoStim lists in the first and second halves of the session.

ECoG/iEEG Analysis

During stimulation lists, EEG data are analyzed to determine when to turn on the stimulator. The raw EEG data is converted from the default referential recording to a bipolar reference between neighboring electrodes as determined by the bipolar montage stored in each patient's '*Talairach structure*'[1]. These bipolar recordings are then filtered to remove 60Hz line noise using a Butterworth stop band filter (58-62Hz range, fourth order). A Morlet wavelet transformation is then applied to the signal (wave number = 5) to estimate spectral power during word encoding periods (8 wavelets log-spaced between 3-180Hz; center frequencies: [3.0, 5.4, 9.7, 17.4, 31.1, 55.9, 100.3, 180] Hz). For analysis of both the training (FR1) and closed-loop data (FR3) we use power from 0-1365 ms after word onset with a mirrored buffer of the same length pre- and post- (1365 ms buffers; total required data window = 4095 ms). Given our lowest frequency wavelet (3 Hz), the minimum required buffer size to avoid convolution edge artifacts is given by one half of the length of the wavelet = 929 ms.[2]

After applying the wavelet transformation we log-transform raw power values, and then downsample the log-transformed power at a 50Hz sampling frequency, prior to removal of the buffer. These log transformed power values are then z-transformed to normalize them based on the within-session mean and standard deviation across all word encoding events in the session. The resulting z-transformed log power values are then features for the pattern classifier described below.

L2-logistic regression classifier: Prior to the start of FR3 we will have fit a regression model to data from multiple FR1 sessions as described below. This regression model will determine the timing of stimulation in FR3 stim lists. Using the spectral features described above we will estimate an L2-penalized logistic regression model to predict whether an item will be subsequently recalled (a proxy for good memory). Before estimating the beta weights (coefficients) on the input features one must specify a penalty parameter to be used in the estimation routine. We determine the penalty parameter based on the penalty value that leads to the highest average AUC in all prior RAM subjects with at least two FR1 sessions. The penalty parameter will be inversely weighted according to the class imbalance (Hastie et al., 2001). Class weights for each class are computed as $(1/n_A)/((1/n_A + 1/n_B)/2)$ where n_A is the number of events class A events, and n_B is the number of class B events. This ensures best fit

[1] *Talairach structure* refers to the eeg_toolbox/PTSA data object used by the RAM team to represent information about each patient's montage. For the purposes of FR3, the relevant information is the set of bipolar electrode pairs, not the coordinate system used to represent the electrode spatial locations. The structure in fact contains electrode information in Montreal Neurlogical Institute (MNI) space; the 'Talairach' nomenclature has nonetheless been maintained from earlier versions of the toolboxes.

[2] One half of the length of the wavelet is given by $n_z s f_s$ where n_z = number of standard units on the Gaussian; s = standard deviation of the Gaussian wavelet window; and f_s = sampling frequency.

values of the cost parameter, C, are comparable across different class distributions. For example, if the proportion of observations in classes A/B are .25 and .75 respectively, then the ratio of the weights for the penalty parameters for each class will be 3:1. The weight for class A will be .25/((.25+.75)/2) = .5, and the weight for class B will be .75/(.25+.75)/2 = 1.5. As new participants are collected and analyzed, their best penalty parameter will be added to the distribution of all previous patients.

Criterion for progressing from FR1 to FR3. Each patients classifier will be tested for out-of-sample performance using leave-one-session-out (LOSO) cross validation. The true LOSO AUC will be compared to a AUC distribution generated (200 samples) from classification of label-permuted input data. $P < 0.05$ one-tailed will be used as the significance threshold for continuing to FR3. At least three sessions with a minimum of 15 lists per session are required. If classifier is not significant with three sessions, additional FR1 sessions should be collected.

Closed-loop Procedure

During the item presentation phase of the StimLists in FR3 we will evaluate the classifier output during the 200ms-1350ms interval following word onset to decide whether to apply stimulation during the next item presentation (With the 950ms buffer on each end, this means an interval of length 3*1365 = **4095 ms**). If the classifier output during this interval is below a threshold value we will stimulate the subsequent word. We will use Youden's *J* statistic to determine the threshold for stimulation individually for each patient. This statistic is given by *J* = Sensitivity + Specificity – 1 and can be represented visually as the height of the ROC curve above the unity (chance) line. We select the maximal *J* value across all points in the patient's FR1 ROC curve as our threshold for stimulation (Schisterman et al., 2005). This is equivalent to the threshold that maximizes classifier performance for detecting true positives while minimizing false alarms.

Data Normalization Process. The values that are input to the real-time classifier will be normalized to ensure the range of the data match the range expected by the classifier model. This will be done using the mean and standard deviation across word encoding events in the first four word lists (one practice list and three NoStim lists). Mean and standard deviation estimates will be updated as more data are collected. Updates will occur after every NoStim list (following the recall period). To do so, StimControl.m will need state information about timing of word onset.

Classifier Validation. To validate the classifier for use in FR3, the output from the classifier for each item presented on the first four lists will be compared to the classifier output when tested on data from all previous sessions using a two-sample Kolmogorov-Smirnov test. The null hypothesis that the FR3 and training data come from the same distribution must not be rejected (i.e., $P > 0.05$) for the FR3 task to continue. If this hypothesis is rejected, an error will be thrown and the session will proceed as an FR1 session. StimControl.m will also output (to the command line and to a log file on disk) the results of the two-sample Kolmogorov-Smirnov test comparing the distributions.

Stimulation Artifacts. We will avoid decoding brain data during stimulation. We will accomplish this by decoding a word presentation and stimulating on the following word (see *Stimulation Onset Timing* below). Using this approach, when a stimulation decision is made, 500 ms of stimulation will be delivered at the next word onset, leaving 1100ms of word encoding + at least 750ms of ISI until the next word presentation, when the next classifier decoding would occur.

Stimulation Onset Timing. A stimulation waveform, specified by the PS-defined parameters (see below), will be triggered when the multivariate classifier detects that the classifier output is below threshold. Specifically, stimulation decisions will be made based on classifier decoding of the word presentation interval. If the classifier output for the decoded word falls below threshold, a stimulation waveform will be triggered to coincide with onset of the *next* word in the list.

Selection of Closed Loop Stimulation Parameters

We will use the *J* statistic from all of a given patient's FR1 sessions to determine threshold (*K*) for identifying poor memory states in all PS2 and PS3 task sessions (e.g., bottom 30th percentile, etc.). Across all such-identified poor memory states we will find the parameter set (\vec{theta}) across all PS sessions that maximizes the t-statistic comparison between post-pre classifier output and the FR1 control comparison set. Based on the parameter manipulated across PS2 and PS3 tasks we expect to identify a set theta that includes:

- Stimulation location (bipolar electrode pair)
- Amplitude **(SafeAmp-.5, SafeAmp-.25 or SafeAmp mA)**
- Pulse frequency **(10, 25, 50, 100, 200 Hz or single pulse)**
- Theta pattern frequency **(3, 4, 5, 6, 7, 8 Hz or n/a)**
- Theta train frequency **(100, 200 Hz or n/a)**

In the absence of patient-specific PS data, theta will be chosen according to the best group-average parameters from the vault of PS datasets in existence at the time of the patient's participation. *K* will be determined based on the patient's own FR1 data across all sessions using the J-statistic as above.

Other Stimulation Parameters:

- Based on the aggregate PS1 data pulse duration will be set to 500ms
- Bipolar stimulation
- Pulse width: 600 microsecond pulse (300 us per phase)

References

- Hastie. T., Tibishirani, R., & Friedman, J. (2001). The elements of statistical learning: data mining, inference and prediction. *New York: Springer*.
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